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Multimedia Communications

Quality of Experience for Visual Multimedia Services (Images and Videos)

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Part of the slides courtesy of prof. Federica Battisti

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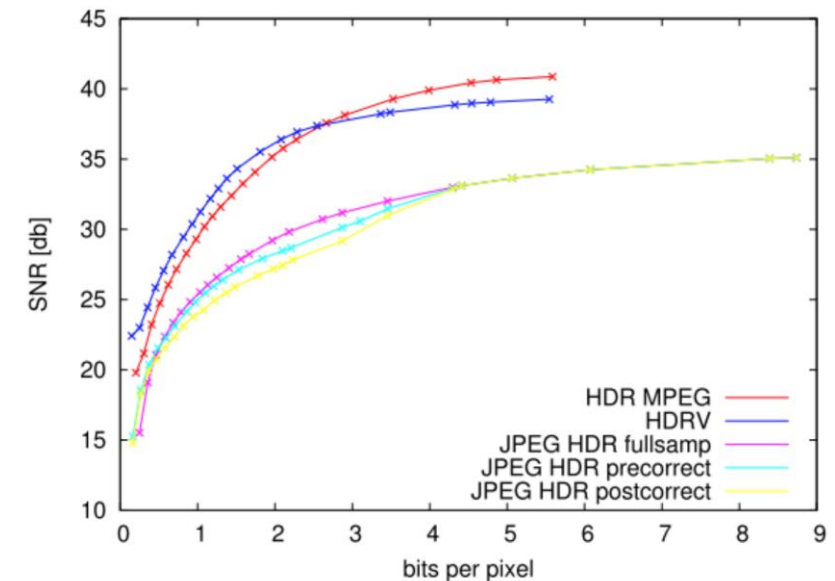


- Introduction to Quality Assessment in Multimedia Services
- Subjective Quality Assessment
- Objective Quality Assessment

- The purpose of Quality assessment in the context of digital multimedia services is to associate a quantitative grade to an image, a video, a sound, or any other possible *stimulus* i.e., a **signal** that can be perceived by a **human observer**
- Since a stimulus is a *signal*, quality assessment can be seen as a *functional*, i.e. an operator associating a number to a signal / several signals
- In this case we are performing an **objective quality assessment**
- However, what finally counts is the impact on the observers: quality assessment can be performed by asking people to evaluate the stimuli
- In this case we are performing a **subjective quality assessment**

Why is QA needed?

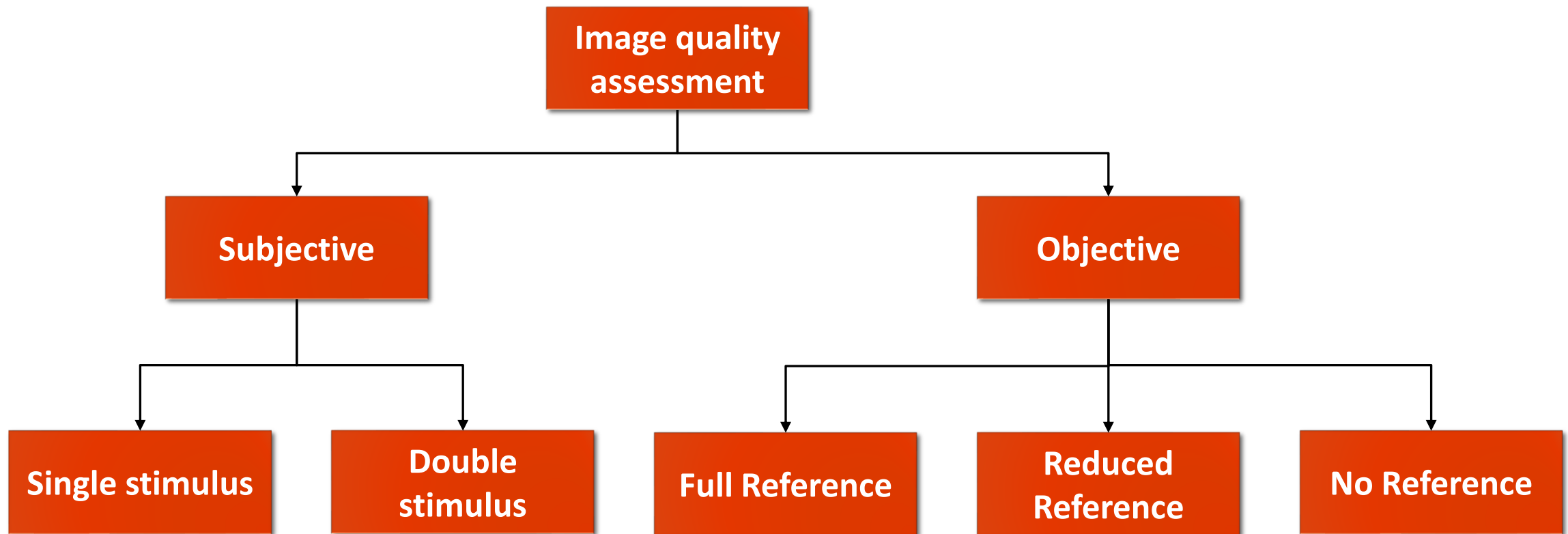
- Quality assessment plays a major role in the design of Multimedia services: it is the “benefit” term in the cost-benefit analysis of a service: How good is the “quality” of the delivered signal?
- The cost depends on the application
- In the case of compression / transmission systems, the cost is typically the coding rate
- But it can also include the number of operations (time complexity) and the amount of memory (space complexity) required





- The trade-off between coding rate and quality is of central importance in our context
- While the coding rate (and, to some extent, the complexity) are easy-to-measure quantities, quality is a whole different game
- In this class we will present the different ways to assess the quality of images and videos
- Mastering the Quality / Rate trade-off is needed in order to make decision in the design of a Multimedia Service systems
 - E.g.: how to compare two compression methods? how to set up a streaming service? etc.

Taxonomy of quality assessment methods





Subjective quality evaluation

- PRO: The most reliable way to measure the quality of experience of multimedia services.
- CON: Subjective tests are usually costly, and time-consuming and require many users to provide reliable results.
- To obtain reliable and reproducible subjective results, it is fundamental to define and describe each of the following elements very carefully:
 - **Laboratory equipment**, including screen and viewing distance, illumination and reverberation characteristics of the room.
 - **Data set**, including the original and processed contents and their distribution
 - **Test methodology**, including the rating target (quality, comparison, or impairment), the scale (categorical or continuous) and the type of stimuli (single or double).
 - **Score processing**, including score normalization, outlier detection, mean score and confidence intervals computation, and significance tests.

- General viewing conditions (room illumination, chromaticity of background, peak luminance, monitor contrast ratio, ratio of luminance of background behind picture monitor to peak luminance of picture)
- Standardized conditions:
 - Viewing distance
 - Observation angle
 - Room environment-colour scheme





Example of Viewing distances

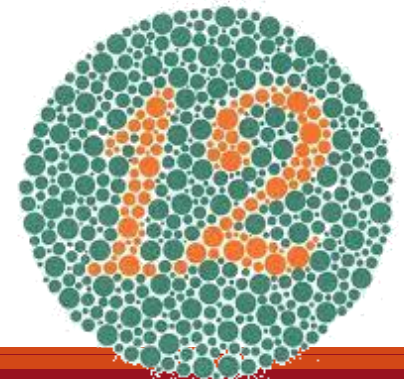
Image system	Reference	Aspect ratio	Pixel aspect ratio	Optimal horizontal viewing angle	Optimal viewing distance
720 × 483	ITU-R BT.601	4:3	0.89	11°	7 H
640 × 480	VGA	4:3	1	11°	7 H
720 × 576	ITU-R BT.601	4:3	1.07	13°	6 H
1024 × 768	XGA	4:3	1	17°	4.5 H
1280 × 720	ITU-R BT.1543 and BT.1874	16:9	1	21°	4.8 H
1400 × 1050	SXGA+	4:3	1	23°	3.3 H
1920 × 1080	ITU-R BT.709	16:9	1	31°	3.2 H
3840 × 2160	ITU-R BT.2020	16:9	1	58°	1.6 H
7680 × 4320	ITU-R BT.2020	16:9	1	96°	0.8 H

H: picture heights

- Using monitors with different characteristics will yield different subjective image qualities
 - Monitor processing
 - Resolution
 - Contrast
 - Brightness
 - Motion artefacts



- Observers may be **expert** or **non-expert** depending on the goals of the assessment
- At least 15 observers should be used
- Prior to a session, the observers should be screened for **normal visual acuity** on the Snellen or Landolt chart, and for **normal color vision** using specially selected charts
- Experimenters should include as much detail as possible on the characteristics of their assessment panels to facilitate further investigation of this factor

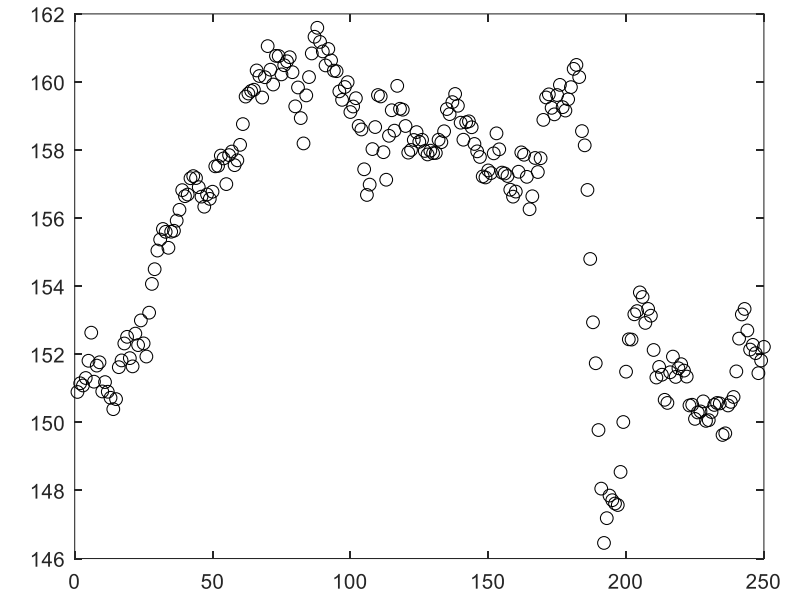
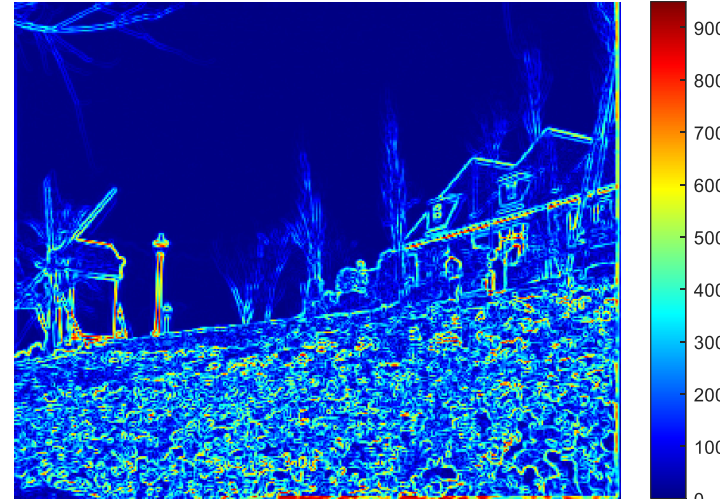


- The selection of the stimuli is largely dependent on the **goal** of the subjective experiment (e.g., comparing compression algorithms, evaluate the performances of an enhancement method, ...)
- For **video stimuli**, it can be useful to compare the relative **Spatial Information** and **Temporal Information** found in the available sequences
- If a small number of test sequences are to be used in a given test, it may be important to choose sequences that span a large portion of the ***spatio-temporal information***

- **Spatial Information (SI):** A measure that indicates the **amount of spatial detail** in a picture. It is usually higher for more spatially complex scenes.
- The SI is based on the Sobel filter, i.e. a discrete analog of the spatial gradient
- The output of a Sobel Filter is large near object contours and small in flat areas
- Each video frame at time n (F_n) is first filtered with the Sobel filter [$Sobel(F_n)$]
- The standard deviation over the pixels (σ_{space}) in each Sobel filtered frame is then computed. This operation is repeated for each frame in the video sequence and results in a time series of SI of the scene.
- The maximum value in the time series (max_{time}) is chosen to represent the SI content of the scene. This process can be represented in equation form as:

$$SI = max_{time}\{\sigma_{space}[Sobel(F_n)]\}$$

Example of Sobel Filtering and SI



An image, the result of Sobel filtering and the temporal behavior of the standard deviation of filtered images
SI=161.6 for this video sequence

- **Temporal information (TI)**: a measure that indicates the amount of temporal changes of a video sequence. It is usually large for high-motion sequences
- TI is based upon the motion difference feature, $M_n(i, j)$, that is the difference between the pixel values at the same location in space but at successive times or frames. $M_n(i, j)$ as a function of time (n) is defined as:

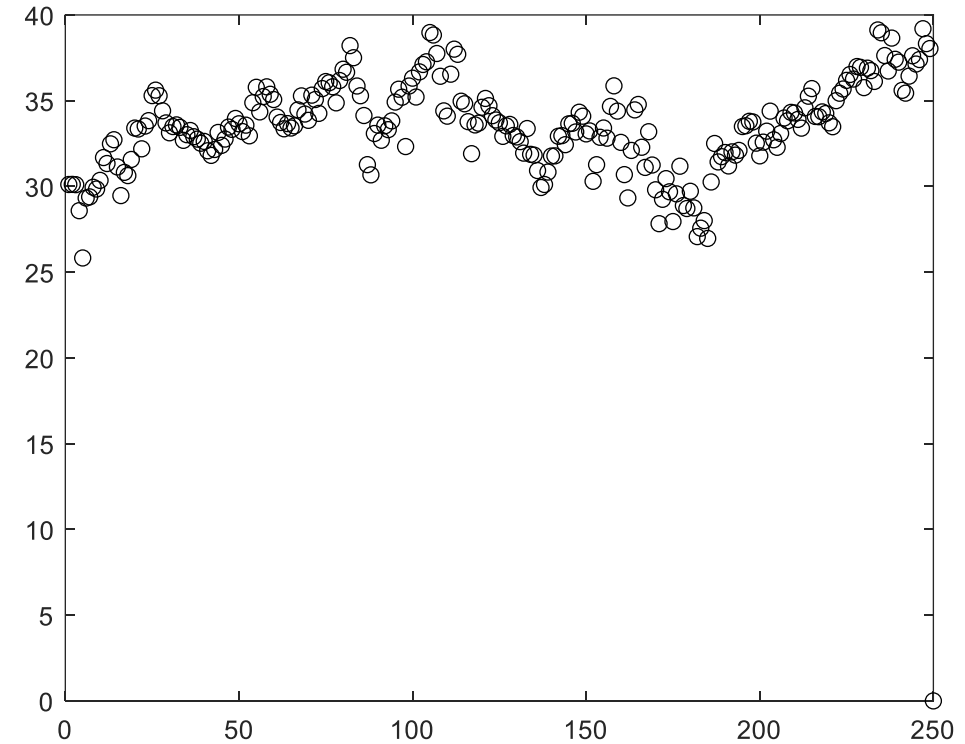
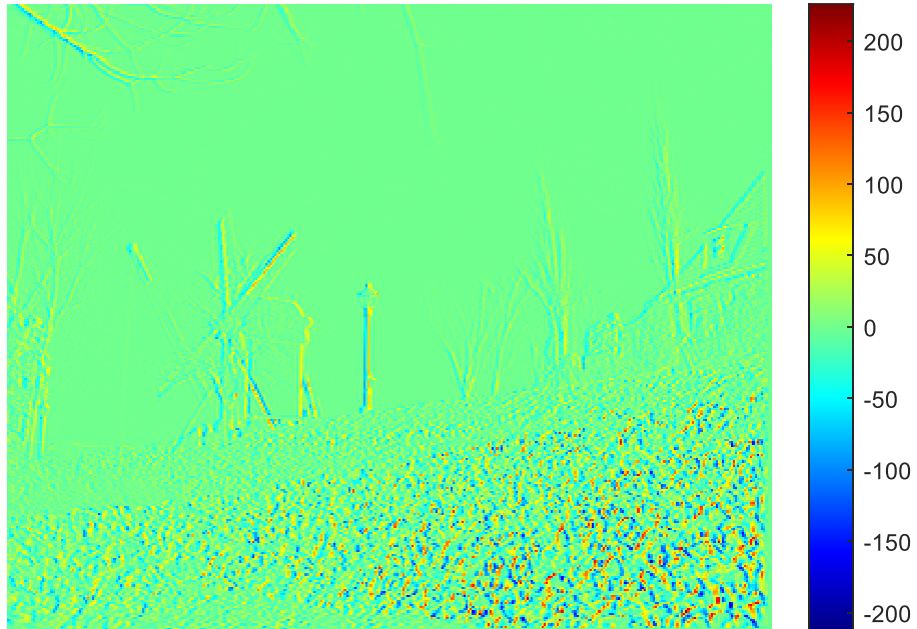
$$M_n(i, j) = F_n(i, j) - F_{n-1}(i, j)$$

here $F_n(i, j)$ is the pixel luminance at the i^{th} row and j^{th} column of n^{th} frame in time. The TI measure is computed as the maximum over time ($\max_{time}$) of the SD over space (σ_{space}) of $M_n(i, j)$ over all i and j :

$$TI = \max_{time} \{ \sigma_{space} [M_n(i, j)] \}$$

Diverse motion in adjacent frames will result in higher values of TI .

Temporal Information

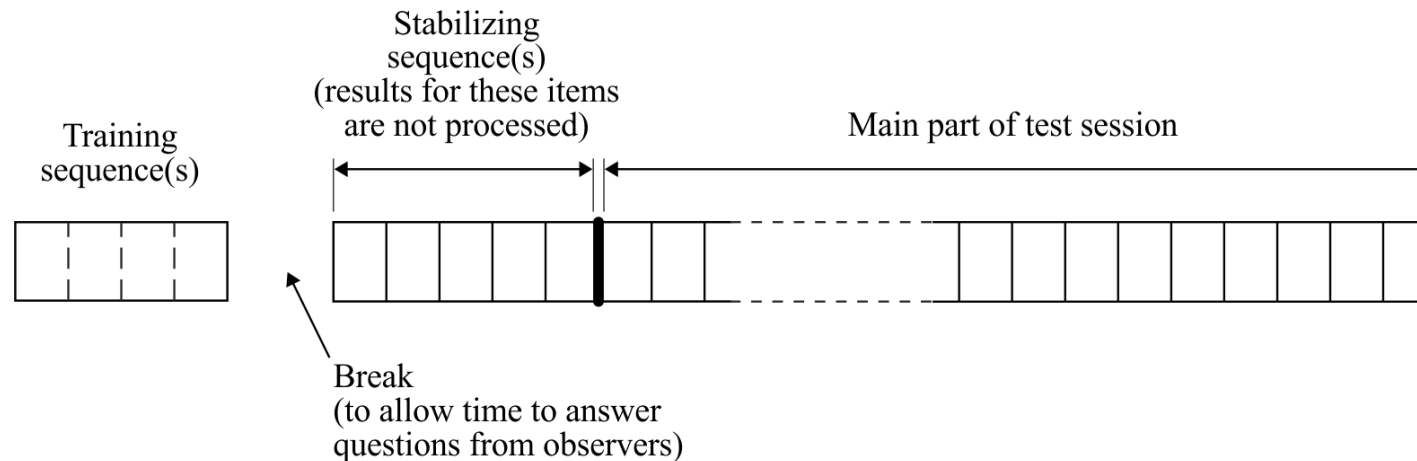


The Motion Difference for a given n and the behavior of its standard deviation
TI = 39.20

Instructions for the assessment

- Assessors should be **carefully introduced** to the method of assessment, the types of impairment or quality factors likely to occur, the grading scale, the sequence and timing
- Training sequences demonstrating the range and the type of the impairments to be assessed should be used with illustrating images other than those used in the test, but of comparable sensitivity.

- A session should not last more than **half an hour**
- At the beginning of the first session, about five “dummy presentations” should be introduced to stabilize the observers’ opinion. The data issued from these presentations must not be considered in the results of the test
- The images and impairments should be presented in a pseudo-random sequence and, preferably in a different sequence for each session



- Participants may
 - change their criteria,
 - become more sensitive (training),
 - become less sensitive (tiredness)
- To control the learning effect
 - Run training session that should cover the range of stimuli so that a participant can determine his/her criteria
 - Keep the sessions short (<20 min, <40 min)
 - Pay participants
 - Randomize stimuli (as much as possible)

Single-stimulus (SS) or Absolute category rating (ACR)

- In SS methods, a single image or sequence of images is presented and the assessor provides an index of the entire presentation.
- The test material might include only test sequences or both the test sequences and their corresponding reference sequence.
- In the latter case, the reference sequence is presented as a freestanding stimulus for rating like any other test stimulus.



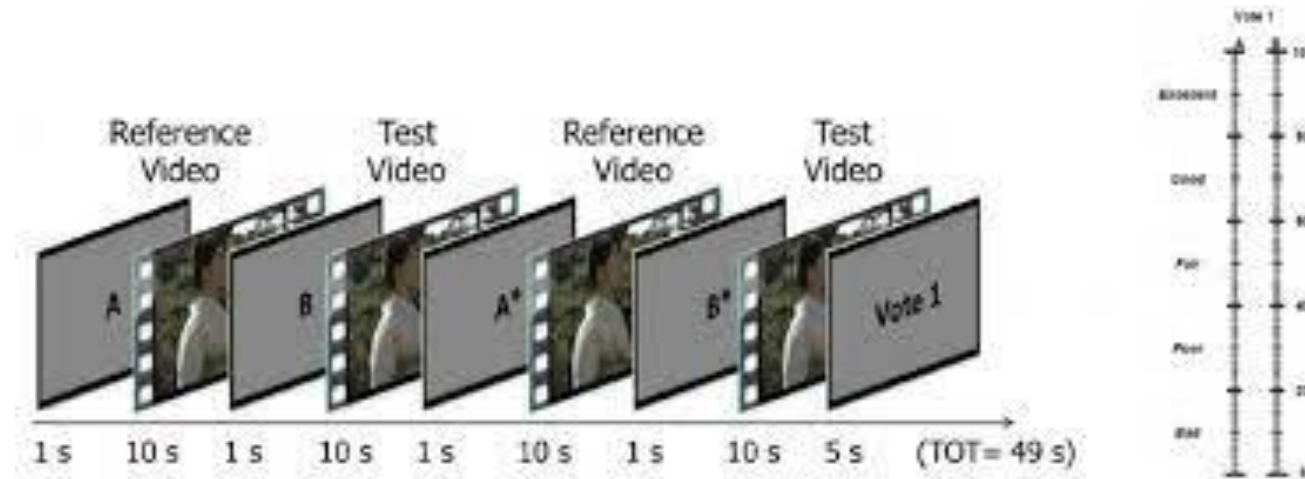


Absolute category rating with hidden reference (ACR-HR)

- The test sequences are presented one at a time and are rated independently on a category scale. The test procedure **must include** a reference version of each test sequence shown as any other test stimulus. This is termed a hidden reference condition.
- During the data analysis, a differential mean opinion score (DMOS) of quality will be computed between each test sequence and its corresponding (hidden) reference

Double Stimulus Impairment Scale (DSIS)

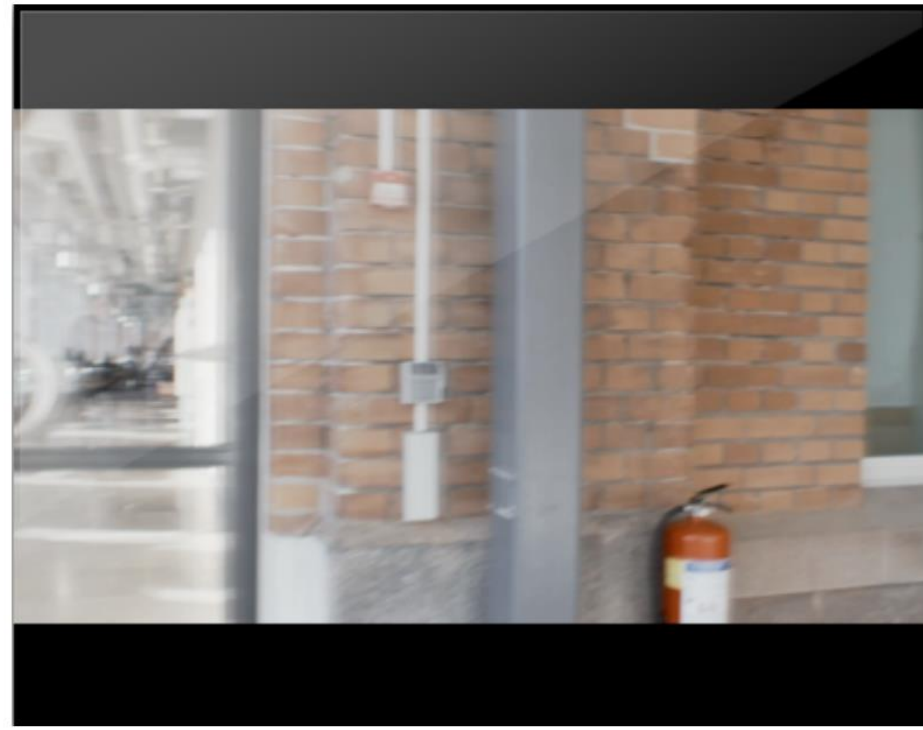
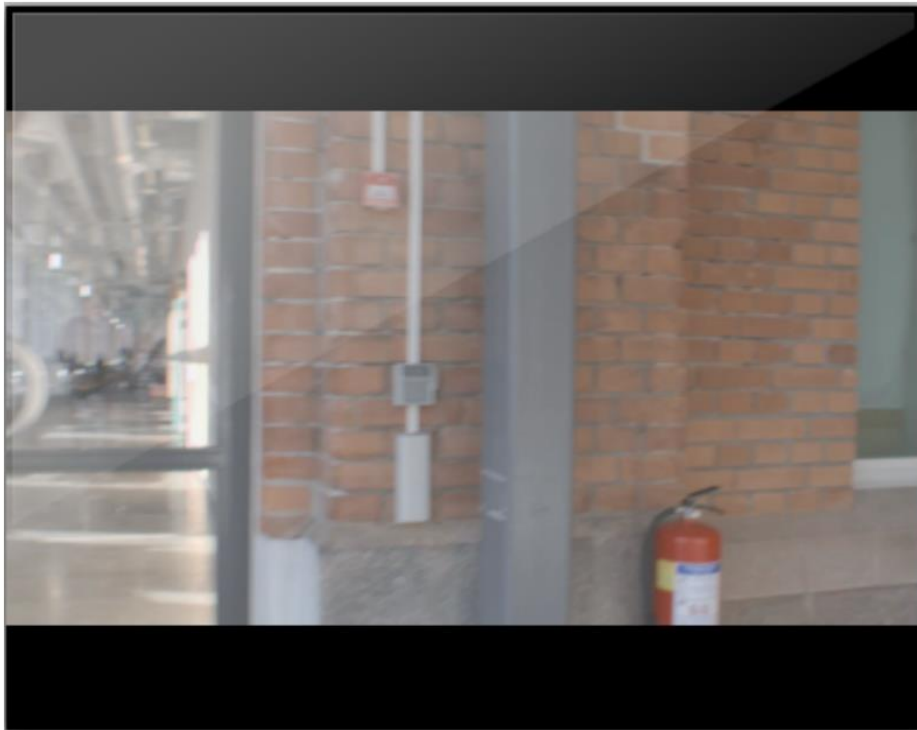
- Reference and processed sequence/image are shown
- Viewers rate degradation** on discrete scale: imperceptible, perceptible but not annoying, fair, poor, bad



- The assessor is first presented with an unimpaired reference, then with the same image impaired. Following this, he is asked to vote on the second, keeping in mind the first.
- The assessor is presented with a series of images or sequences in random order and with random impairments covering all required combinations.

Pair-wise comparison (PC) method

- Example: video quality ranking among n sequences
- Task: You will see two video sequences one after another. Select the sequence of higher quality.

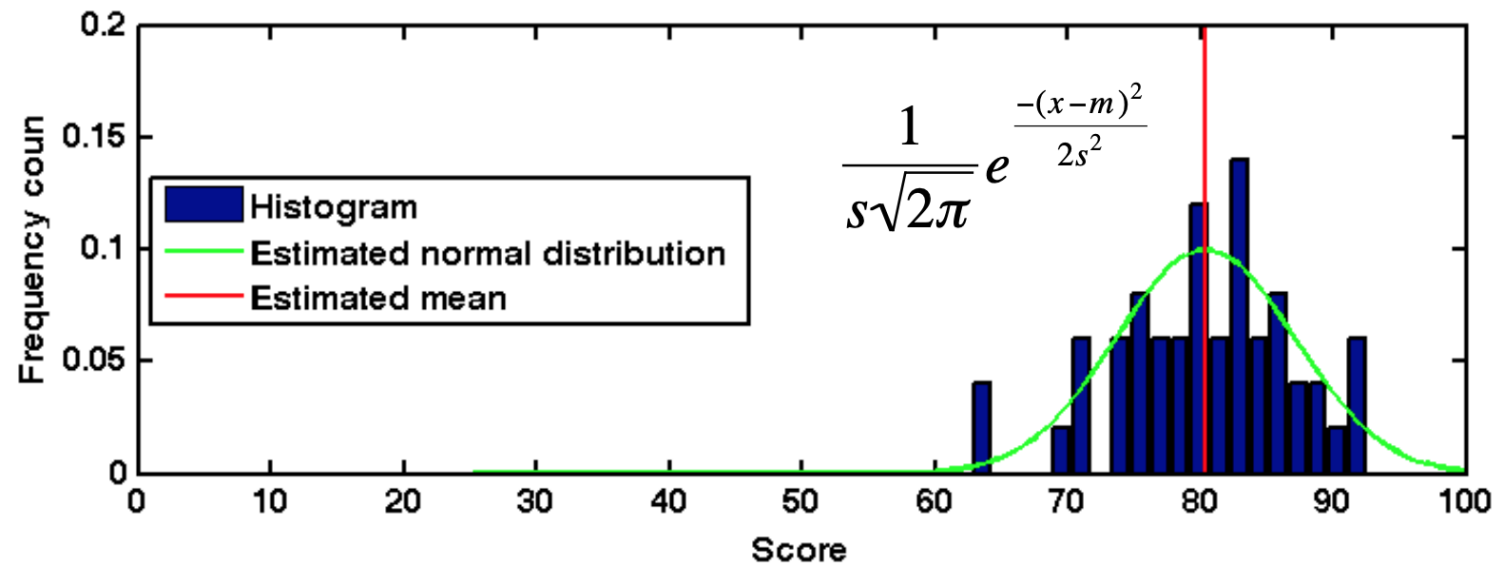


- Results are stored in a **comparison matrix**

$$C = \begin{matrix} & \begin{matrix} c1 & c2 & c3 \end{matrix} \\ \begin{matrix} c1 \\ c2 \\ c3 \end{matrix} & \begin{bmatrix} 0 & 3 & 1 \\ 3 & 0 & 1 \\ 1 & 3 & 0 \end{bmatrix} \end{matrix}$$

- $C_{ij} = n$ means that condition C_j was preferred over C_i n times

- Measured values are considered outcomes of a random process
- Example: 30 observers rated an image from 0 to 100



- Assuming that the measured distribution is normal
- mean – estimates as an average of observer rates

Standard deviation and standard error

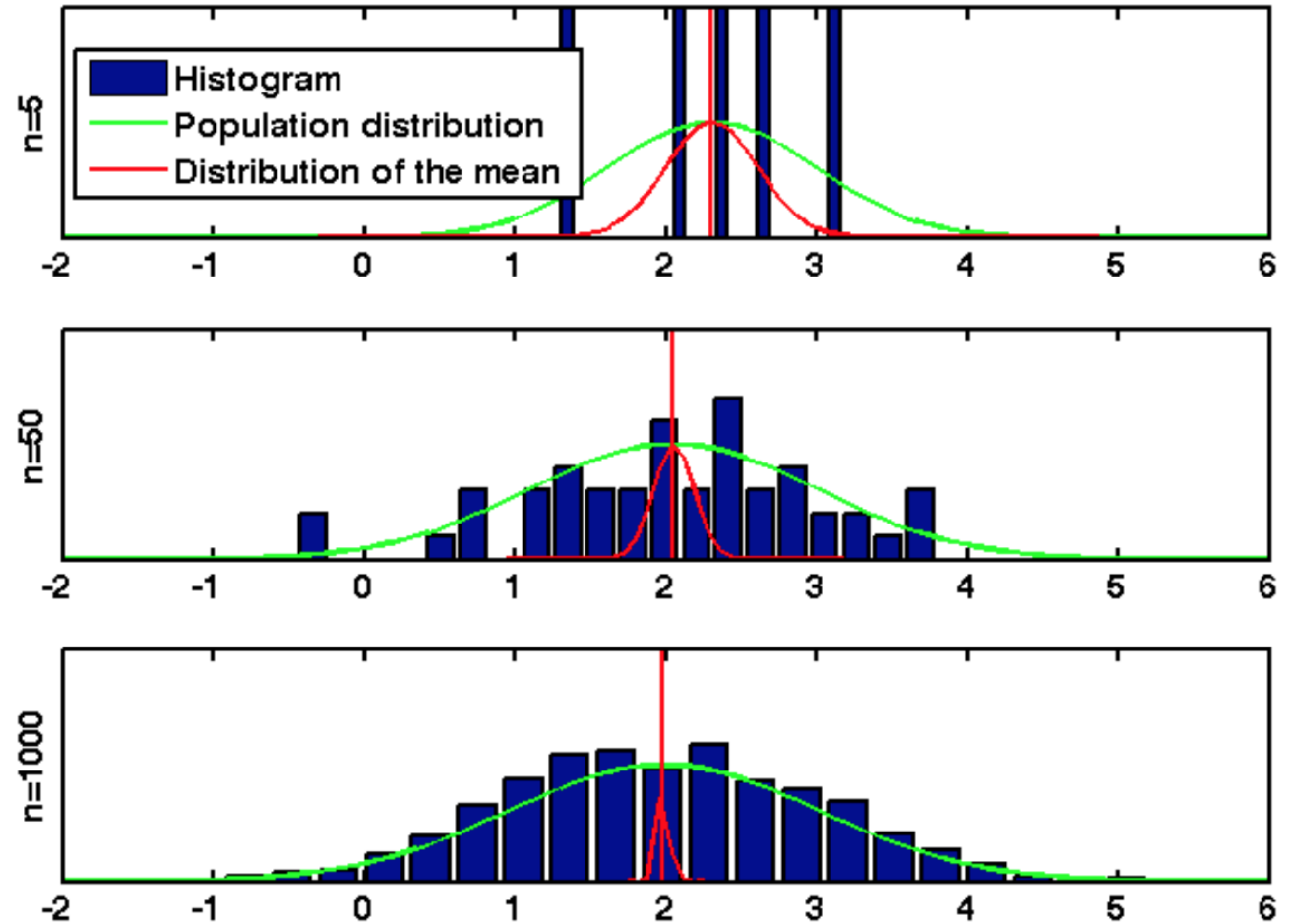
- Standard deviation

$$s = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - m)^2}$$

- Standard error of the mean

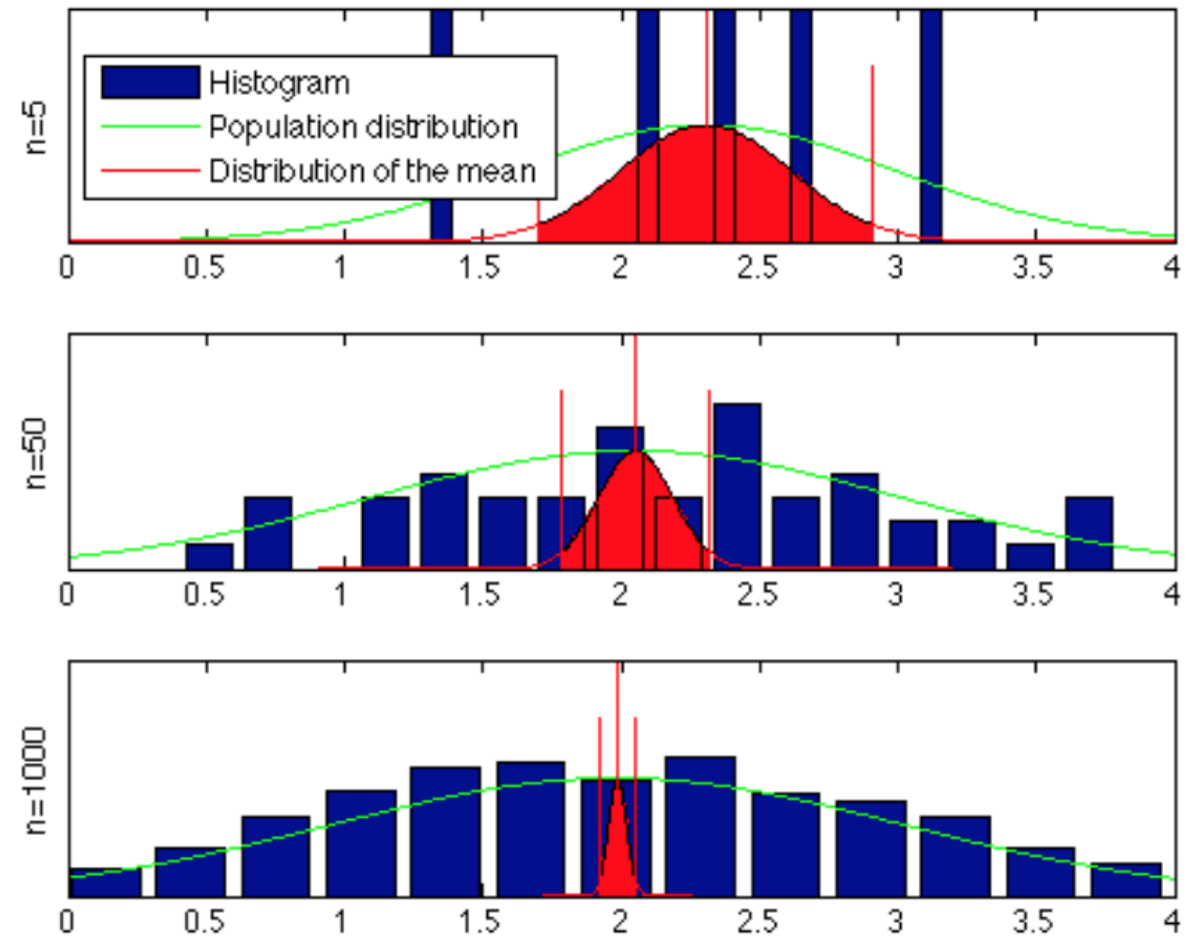
$$SE = \frac{s}{\sqrt{N}}$$

N = sample size, m =mean



- If the same experiment is repeated with the same number of observers, in 95% of the cases the average value is expected to be within the range of the confidence interval

$$ci = [-1,96 \cdot SE; 1,96 \cdot SE]$$



Comparison of the methods

- An important issue in choosing a test method is the fundamental difference between methods that use explicit references (e.g., DSIS), and methods that do not use any explicit reference (e.g., ACR, ACR-HR and PC). This second class of method does not test fidelity.
- The **DSIS** method should be used when testing the fidelity of transmission with respect to the source signal
 - DSIS has long been a key method specified in [ITU-R BT.500-14], for the assessment of TV pictures whose typical quality represents the extreme high levels of videotelephony and videoconferencing.
 - The specific comments of the DSIS scale (imperceptible or perceptible) are valuable when the viewer's detection of impairment is an important factor.
- **ACR** is easy and fast to implement, and the presentation of the stimuli is similar to that of the common use of the systems. Thus, ACR is well suited for qualification tests.

Comparison of the methods

- **ACR-HR** has all the advantages of ACR with respect to presentation and speed.
- The principal merit of ACR-HR over ACR is that the perceptual impact of the reference video can be removed from the subjective scores. This **reduces the impact of scene bias** (e.g., viewers liking or disliking a reference video), **reference video quality** (e.g., small differences in camera quality) and **monitor** (e.g., professional quality versus consumer grade) upon the final scores.
- ACR-HR is well suited to large experiments, provided that all reference videos are at least "good" quality. However, ACR-HR may be insensitive to some impairments that are easily detected by direct differential methods (e.g., DSIS).
- The principal merit of the **PC** method is its **high discriminatory power**, which is of particular value when several of the test items are nearly equal in quality.
- When a large number of items are to be evaluated in the same test, the procedure based on the PC method tends to be lengthy. In such a case, an ACR or DSIS test may be carried out first with a limited number of observers, followed by a PC test solely on those items that have received about the same rating.

- There is the possibility to outsource the quality assessment task to workers from an internet crowd, who evaluate the quality of multimedia files at their leisure, using their own hardware (no controlled environment and test conditions)



- Need for specific processing of the collected scores to remove unreliable voters.

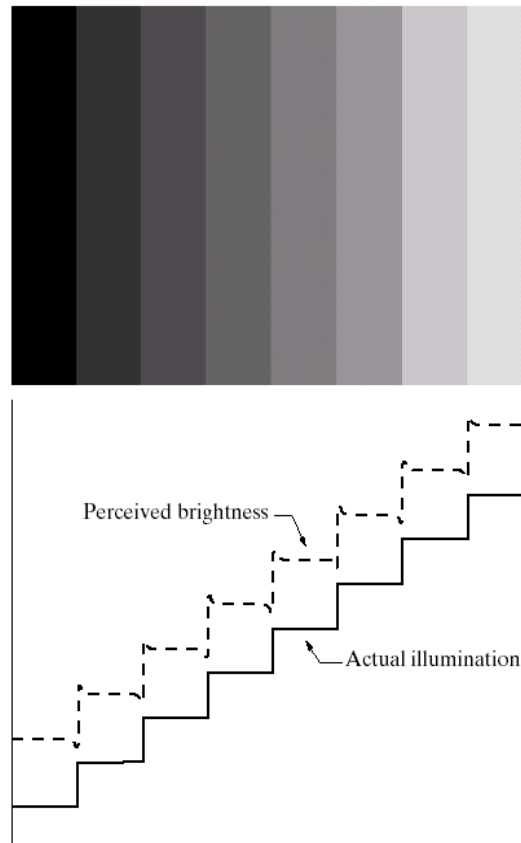


Objective quality evaluation

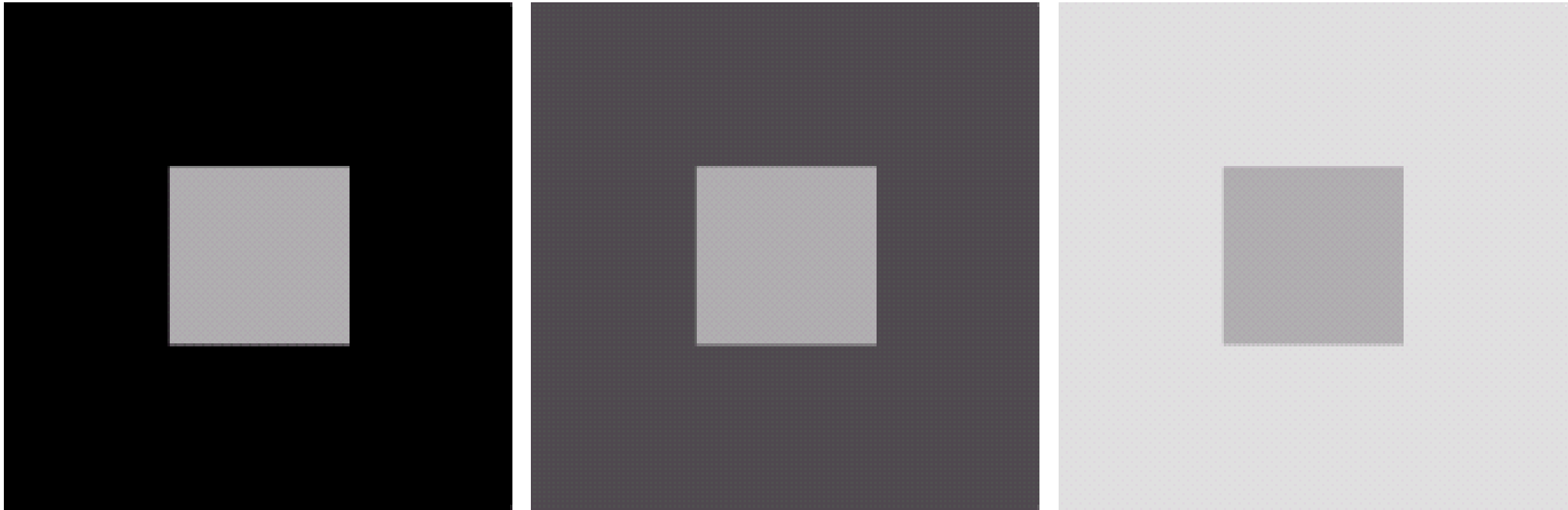
- Quality assessment algorithms usually predict visual quality by comparing a distorted signal against a reference, typically by modeling the human visual system.
- The objective image quality assessment is based on well defined mathematical models that can predict perceived image quality between a distorted image and a reference image.
- These measurement methods consider human visual system characteristics in an attempt to incorporate perceptual aspects in the quality measures.

Perceived Brightness is not a Simple Function of Light Intensity

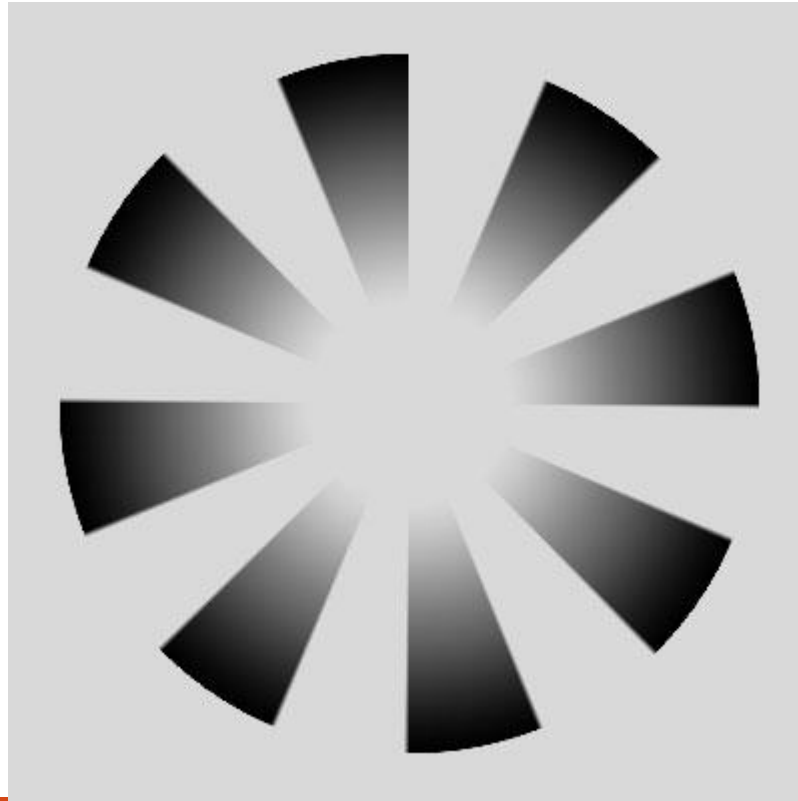
- Mach band pattern: The visual system tends to undershoot or overshoot around the boundaries of regions of different intensities

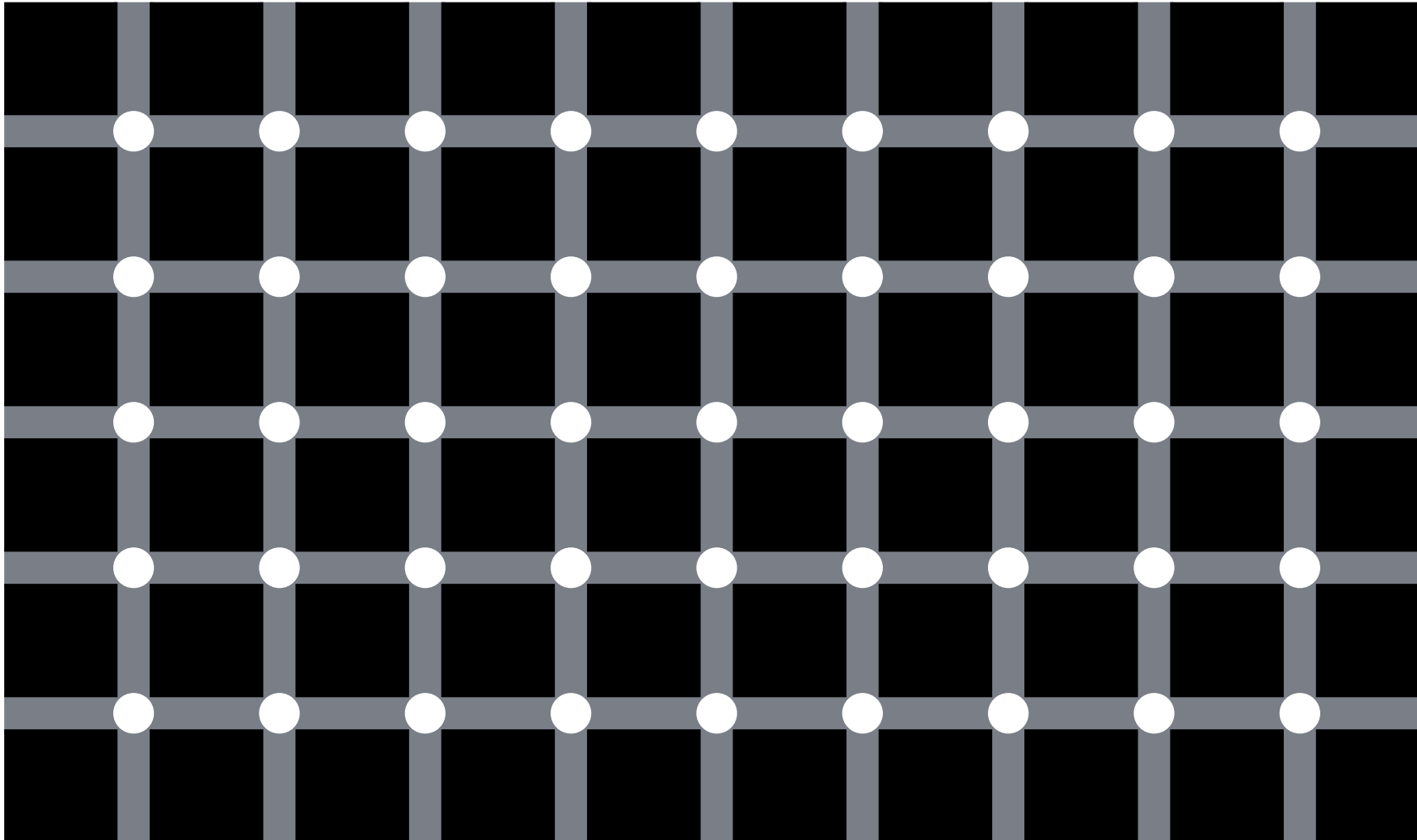


- A region's perceived brightness does not depend simply on his intensity but depends also on the background intensity.

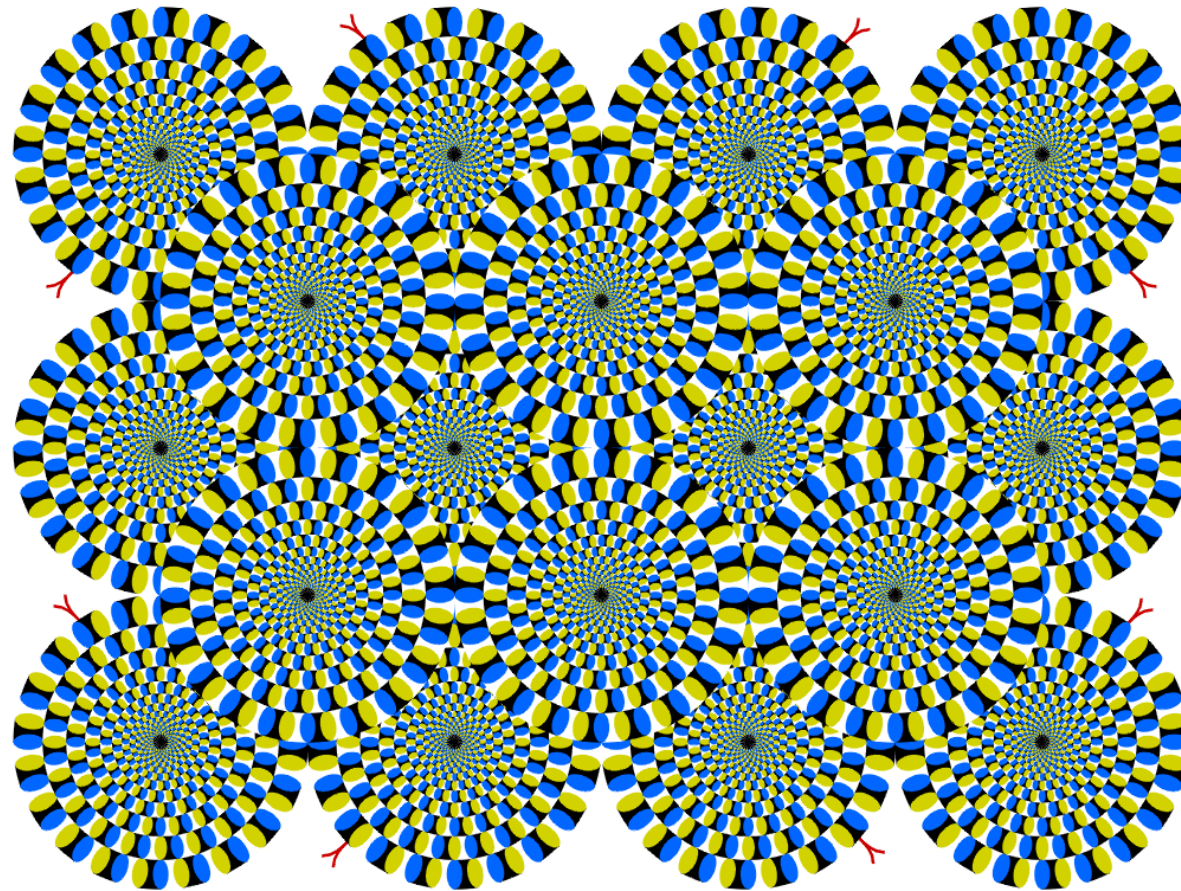


- The HVS fills in nonexisting information or wrongly perceives geometrical properties of objects.





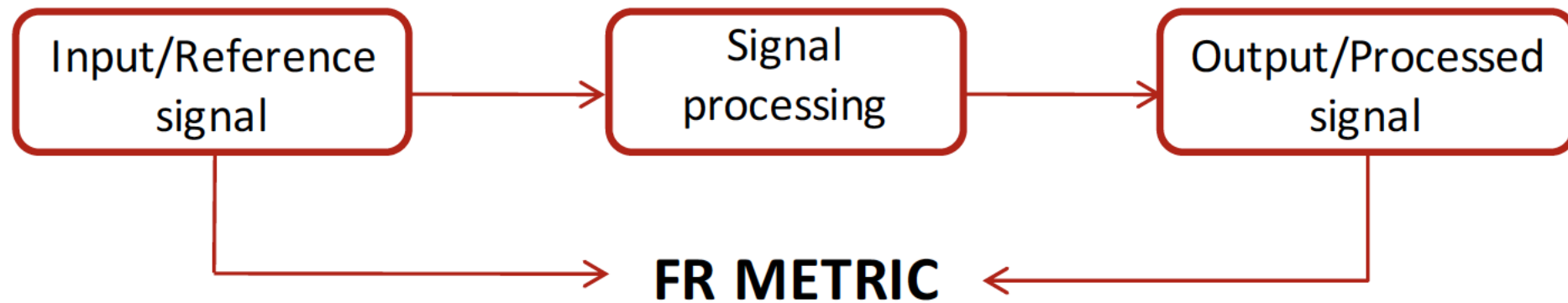
Anomalous Motion Illusion



- Objective algorithms, i.e. metrics, estimating subjective MOS with high level of correlation are desired
 - Full reference metrics
 - No-reference metrics
 - Reduced reference metrics



- Full Reference approach



We start with *full-reference methods*, i.e., quality metrics that depends on both the reference (original) and test (compressed) image

- $I(n, m, c, t)$ is the intensity of component c in position n, m at time t of the reference signal
- The test signal is $\hat{I}(n, m, c, t)$
- The color space is typically YCbCr, but in some cases RGB can be used

- The first image quality metric is the *mean square error* of the luminance component, $\text{MSE}_Y(t)$
- For a single image (image at time t) of a video sequence, we define:

$$\text{MSE}_Y(t) = \frac{1}{NM} \sum_{n,m} \left(I(n, m, 1, t) - \hat{I}(n, m, 1, t) \right)^2$$

- The logarithmic version of the MSE is the PSNR:

$$\text{PSNR}_Y(t) = 10 \log_{10} \frac{255^2}{\text{MSE}_Y(t)} \approx 48 \text{dB} - 10 \log_{10} \text{MSE}_Y(t)$$

- Similarly, one can define $\text{MSE}_{\text{Cb}}(t)$, $\text{MSE}_{\text{Cr}}(t)$, $\text{PSNR}_{\text{Cb}}(t)$, $\text{PSNR}_{\text{Cr}}(t)$
- For example,

$$\text{MSE}_{\text{Cb}}(t) = \frac{1}{\frac{N}{2} \frac{M}{2}} \sum_{n,m} \left(I(n, m, 2, t) - \hat{I}(n, m, 2, t) \right)^2$$

A global quality measure for color images is:

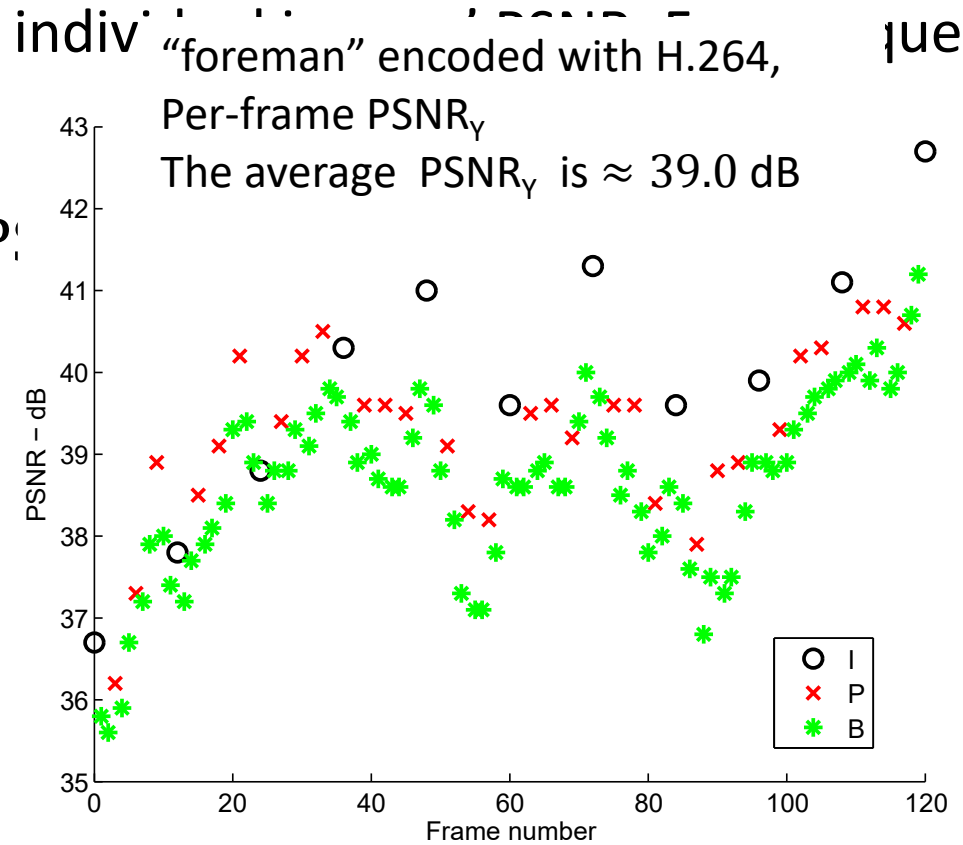
$$\text{PSNR}_{\text{YCbCr}}(t) = \frac{3}{4} \text{PSNR}_{\text{Y}}(t) + \frac{1}{8} \text{PSNR}_{\text{Cb}}(t) + \frac{1}{8} \text{PSNR}_{\text{Cr}}(t)$$

The smaller weights reflect the lesser impact of colors on human perception

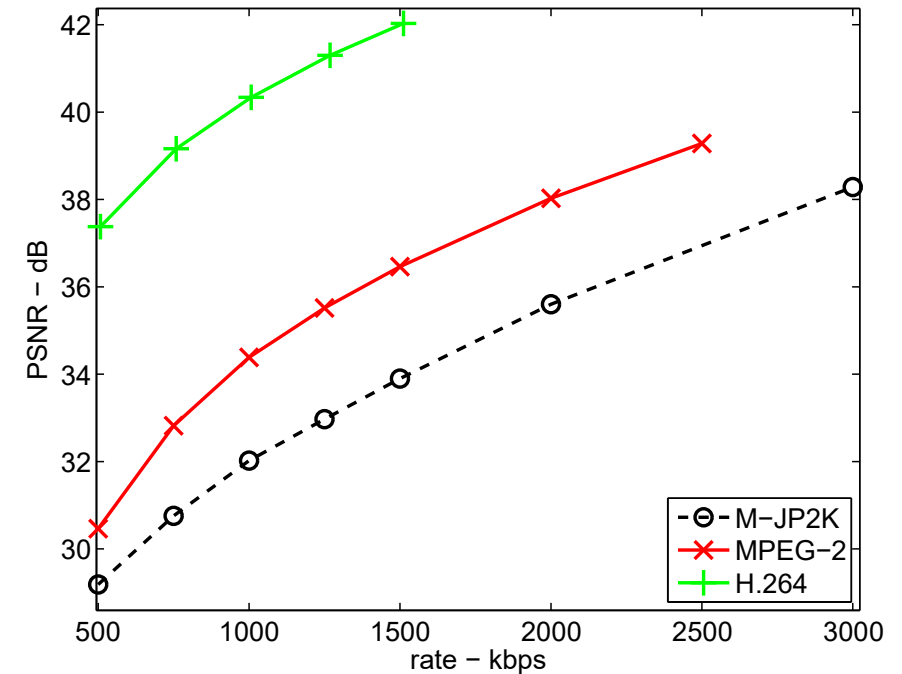
- The PSNR of a video sequence is the average of individual PSNRs of T images

$$\text{PSNR}_Y = \frac{1}{T} \sum_{t=1}^T \text{PSNR}_Y(t)$$

Similarly for PSNR_{Cb} , PSNR_{Cr} , PSNR_{YCbCr}



- Video codec produce rate/distortion curves
- A single-measure comparison can be achieved usign the Bjontegaard metrics:
 1. **Bjontegaard Delta Rate**: it is the average rate difference at the same quality, expressed as %
 - It is computed as the red area divided by the amplitude of the common PSNR interval
 2. **Bjontegaard Delta PSNR**: it is the average PSNR difference at the same rate, expressed in dB
 - It is computed as the blue area divided by the amplitude of the common rate interval



Bjontegaard metrics

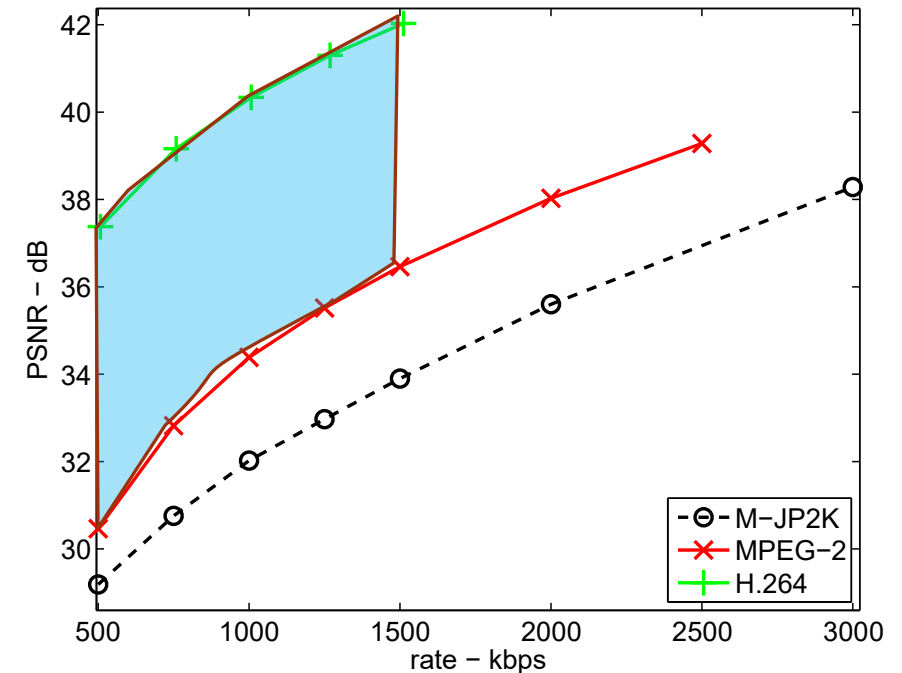
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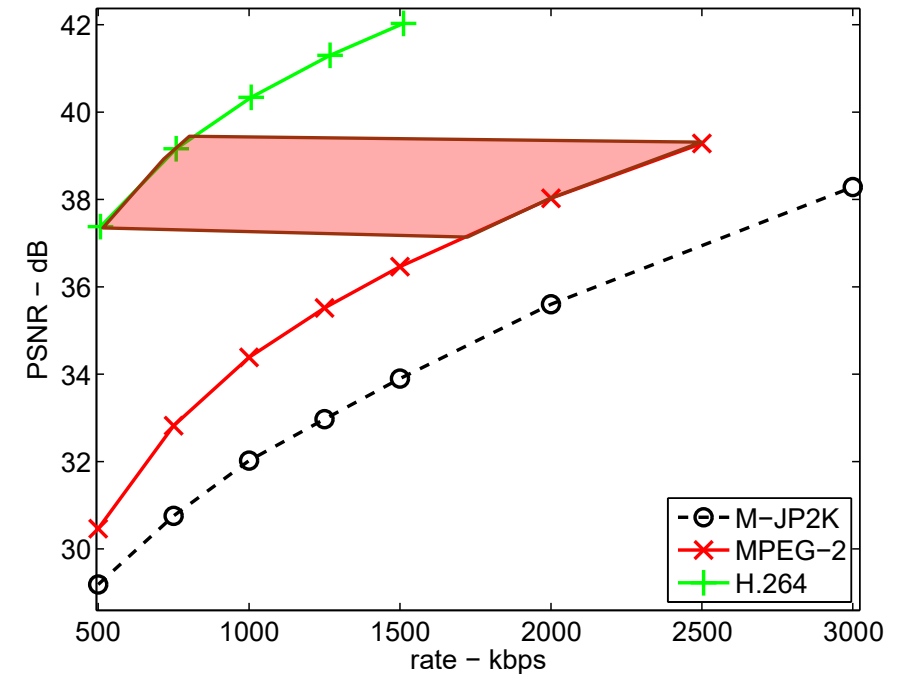
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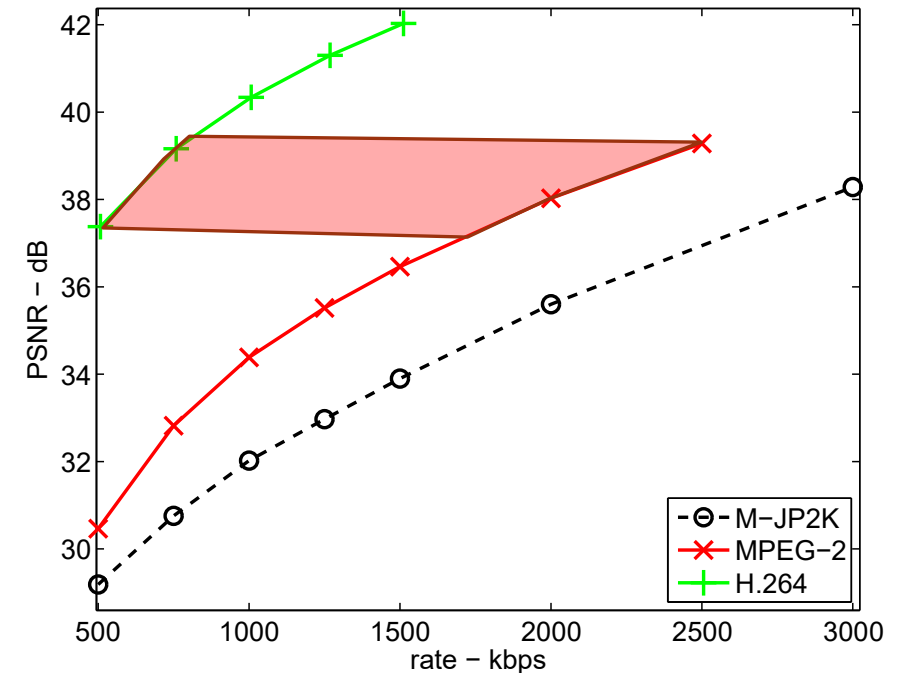
2. **Bjontegaard Delta PSNR**: it is the average PSNR difference at the same rate, expressed in dB

- It is computed as the blue area divided by the amplitude of the common rate interval



Bjontegaard metrics

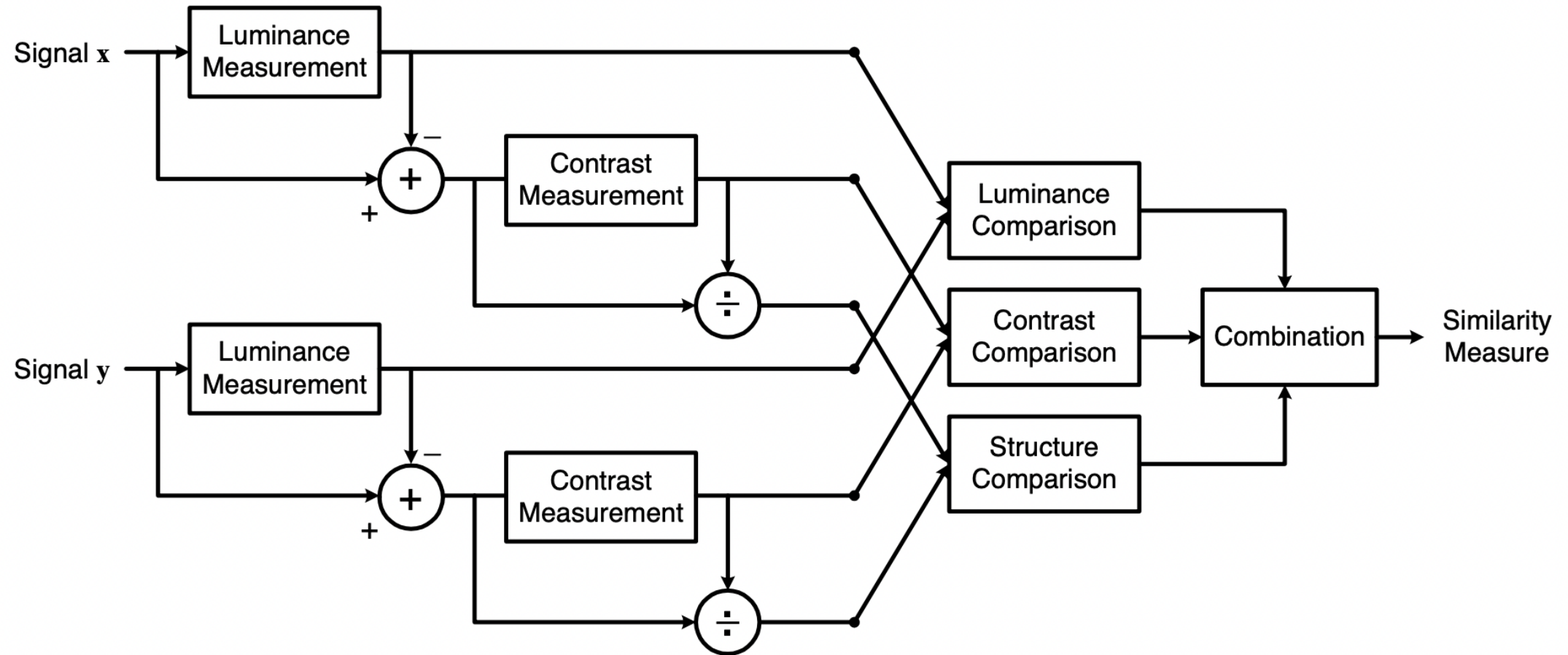
- In practice, the BD metrics are computed as follows:
- For each curve, a small (4 or 5) number of points is considered
- These points are fitted into a parametric representation of the curve (3rd degree polynomial in the variable $t = \log R$)
- The area is computed using the parametric representation of the two curves



SSIM – Structural Similarity Index

- In 2004 a new approach for image quality assessment has been presented, the Structural Similarity Index or SSIM.
- It is a Full reference metric, whose value lies between $[0,1]$ (bad to good)
- The SSIM is designed to improve on traditional metrics like PSNR and MSE, which have proved to be inconsistent with human perception.
- It is based on human visual system.

SSIM – Structural Similarity Index



$$SSIM(x, y) = [l(x, y)]^\alpha [c(x, y)]^\beta [s(x, y)]^\gamma$$

- $l(x, y)$ accounts for luminance change. It is qualitatively consistent with Weber's law, which has been widely used to model light adaptation in the HVS
- $c(x, y)$ accounts for contrast change. It is consistent with the contrast-masking feature of the HVS
- $s(x, y)$ is the structure comparison function
- where α , β and γ are parameters used to adjust the relative importance of the three components

FR: PSNR (↑) and SSIM (↑)

$$PSNR = 10 \times \lg \left(\frac{255^2}{MSE} \right)$$

$$MSE = \frac{1}{M \times N} \sum_{i=1}^N \sum_{j=1}^M [I(i,j) - I'(i,j)]^2$$

PSNR= 20.6 dB

SSIM=0.82



FR: PSNR ($\uparrow$) and SSIM ($\uparrow$)



PSNR= 13.46 dB
SSIM=0.68

Are FR metrics reliable?

Reference image



PSNR = 30.05 dB



Additive Gaussian Noise,
variance 64

PSNR = 25.15 dB



Additive Gaussian Noise,
variance 200

PSNR = 20.42 dB



Additive Gaussian Noise,
variance 600

Are FR metrics reliable?

Reference image

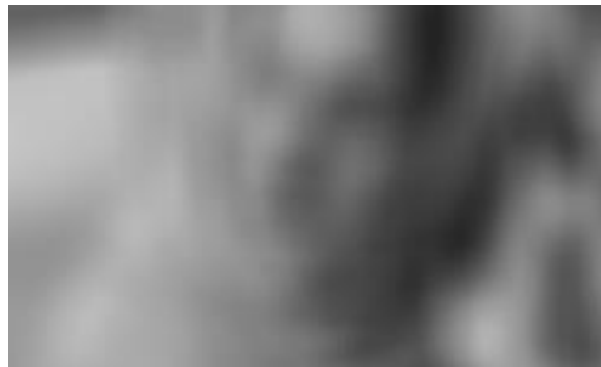


PSNR = 20.42 dB



Additive Gaussian Noise

PSNR = 20.42 dB



Blur

PSNR = 20.76 dB

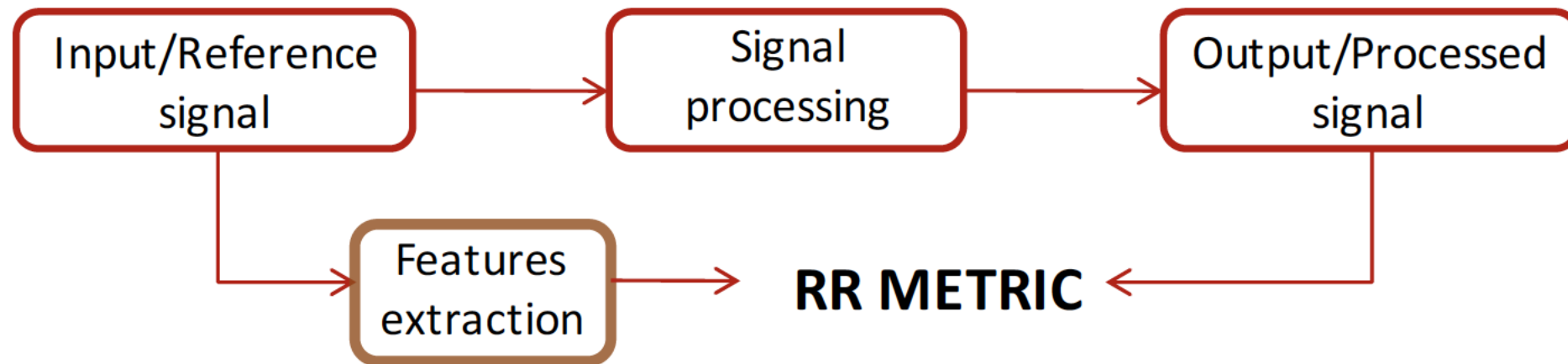


Additive Gaussian Noise
In high frequencies

Why MSE and PSNR are poor?

- MSE and PSNR are widely used because they are simple and easy to calculate and mathematically easy to deal with for optimization purpose
- There are a number of reasons why MSE or PSNR may not correlate well with the human perception of quality
 - Digital pixel values, on which the MSE is typically computed, may not exactly represent the light stimulus entering the eye.
 - Simple error summation, like the one implemented in the MSE formulation, may be markedly different from the way the HVS and the brain arrives at an assessment of the perceived distortion.
 - Two distorted image signals with the same amount of error energy may have very different structure of errors, and hence different perceptual quality

- Reduced Reference approach



- There are many features that have been explored in the past
- The largest part of them focus on a small variety of possible real conditions

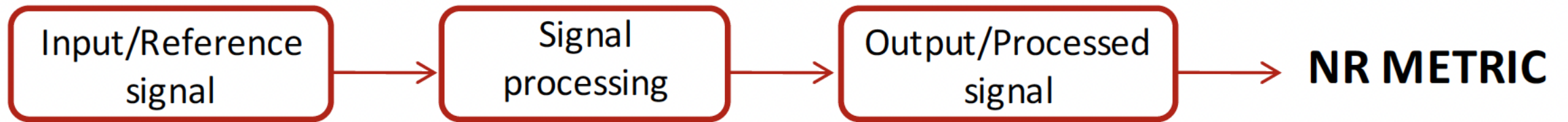


Content type

Distortion type

Distortion levels

- No-Reference approach



Video full reference metrics: VMAF (↑)

- Video Multi-Method Assessment Fusion
- Proposed in June 2016
- Full reference video metric
- VMAF values spans from quality 0–100, where 0 is worst, and 100 is best
- VMAF 93 is a good value to target, but everything with an average over 90 will look good.

How VMAF works

